

ABOUT ME

Accomplished AI Strategist and Leader with over a decade of prolific experience driving innovation across diverse industries. My journey has encompassed roles as a seasoned Data Scientist, adept Software Architect, and influential Trainer, shaping the landscape of energy, security, finance, insurance, medical, and health domains.

My proficiency spans an array of transformative technologies, including Machine Learning, Artificial Intelligence, Image and Video Processing, Data Analytics, and Cloud Computing. At present, my focus lies in spearheading cutting-edge initiatives such as NLP-driven chatbots, generative AI, harnessing the power of Large Language Models (LLMs), predictive analytics, and pioneering computer vision applications.

In addition to my technical acumen, I have honed a remarkable aptitude for assembling and nurturing teams from the ground up, fostering a culture of excellence and innovation. My proven track record in driving projects from conception to fruition, alongside my ability to navigate complex landscapes, makes me a valuable asset for any organization seeking to push the boundaries of technological advancement and drive transformative change.

WORK EXPERIENCE

12/2023 – CURRENT Dharan, Saudi Arabia
DATA SCIENCE CONSULTANT TATA CONSULTANCY (SAUDI ARAMCO)

- Spearheading innovative projects in the realm of state-of-the-art generative AI and natural language processing, focusing on transformative AI solutions within the oil and gas sector.
- Expertly managing the deployment and operation of large language models across local GPU clusters, catering to diverse use cases that require high levels of accuracy and efficiency.
- Pioneering advanced AI techniques including model fine-tuning, Retrieval-Augmented Generation (RAG), and few-shot learning, ensuring they align with stringent data security, privacy, and confidentiality standards.
- Contributing to the strategic development of AI initiatives, driving forward the adoption and integration of generative AI technologies to enhance operational excellence and decision-making processes.

06/2023 – 11/2023
ASSOCIATE SOFTWARE ARCHITECT(ML AND DATA SCIENCE) NORTHBAY SOLUTIONS

In my capacity as an Associate Architect specializing in ML and Data Science, I am deeply immersed in a spectrum of projects focused on NLP, Large Language Models (LLMs), and predictive analytics. My pivotal role involves conceptualizing and contributing to pioneering solutions that leverage the potency of language processing, fine-tuning LLMs, and constructing predictive models to uncover invaluable insights. My commitment to propelling technological frontiers remains steadfast as I collaboratively navigate cross-functional teams to yield profound outcomes.

Furthermore, my expertise extends to leveraging prominent AWS services such as SageMaker and Forecast, which play a pivotal role in amplifying the effectiveness and scalability of our solutions.

10/2021 – 05/2023 Islamabad, Pakistan
DEVELOPMENT HEAD/ ML ARCHITECT TOKTOAI

- 1. Large Language Model (LLM) Enhanced FAQ Chatbot - German Enterprise**
Designed and implemented an advanced chatbot solution to address frequently asked questions pertaining to the company's services. Initially, the chatbot was trained from the ground up, but later it was optimized using a sophisticated Large Language Model (ChatGPT), resulting in significantly improved performance.
- 2. Real Estate lead generation Chatbot**
Engineered an intelligent chatbot tailored for real estate interactions, facilitating user queries and responses. Rigorously tested with a diverse dataset of approximately 1 million user inquiries, the chatbot effectively processed around 20 distinct intents. Notably, it exhibited a nuanced comprehension of contextual queries, catering specifically to the needs of US users.
- 3. Smart Resume Ranking System - UK Recruitment Firm**
Architected an innovative resume ranking solution utilizing state-of-the-art Machine Learning and Natural Language Processing techniques. The system intelligently assessed multiple CVs against an array of job descriptions, employing diverse criteria such as recent work experience, job titles, skills, and qualifications to rank candidates optimally.
- 4. French CV Evaluation System**
Devised a comprehensive resume ranking algorithm focused on French CVs in alignment with job descriptions. Employed sophisticated techniques including Named Entity Recognition, semantic similarity analysis, and segmentation. The solution was elegantly consolidated through Flask APIs, ensuring efficient and seamless integration.
- 5. E-commerce Customized Clothing: Precise Measurement Prediction**

Developed an advanced regression model to predict intricate body measurements like chest size, hip size, and biceps dimensions. Leveraging user-provided inputs such as shoe size, collar size, and height, the algorithm achieved an exceptional R-squared score exceeding 0.98. This comprehensive solution was elegantly orchestrated via Flask and seamlessly deployed on cloud infrastructure.

6. Personal Profile Generation with Transformer Models - NLP Innovation

Engineered an algorithmic marvel for generating personalized candidate profiles utilizing cutting-edge Transformer models. Employing specific keywords as input, the algorithm dynamically crafted compelling summaries of user profiles. This transformative solution showcases the power of modern NLP techniques in producing accurate and succinct profiles.

10/2020 – 09/2021 Islamabad, Pakistan

MACHINE LEARNING TEAM LEAD ASSISTECH SOLUTIONS

1. Employee Performance Insights through NLP and Predictive Analytics

As the lead of the Machine Learning team, I orchestrated an intricate analysis of employee performance by harnessing Natural Language Processing (NLP) and predictive analytics. By meticulously extracting and processing communication data among employees, we conducted a multifaceted evaluation encompassing sentiment analysis, topic modeling, and toxicity detection. Our approach culminated in the categorization of employees into distinct segments including burnout, high performers, and low performers. The comprehensive toolkit involved in this endeavor included Tensorflow, Sklearn, NLTK, Spacy, and Pytorch.

2. Advanced Player Analytics via Computer Vision

In my capacity as the ML team lead, I spearheaded a transformative initiative involving comprehensive player analytics through cutting-edge computer vision techniques. Focusing on softball players, we meticulously computed nuanced angles such as lean forward and backward during crucial actions like set positions, forward motions, lead foot planting, and ball release. Additionally, we accurately determined metrics like ball velocity and center of mass velocity. This endeavor necessitated the fusion of pose estimation, object detection, and classification models, supported by frameworks such as GluonCV, Mxnet, Tensorflow, and openCV.

08/2017 – 10/2020 Islamabad, Pakistan

PRINCIPAL MACHINE LEARNING ENGINEER AND TECHNICAL HEAD EVEATI PVT. LTD.

1. Superstore Customer Insights through Innovative Computer Vision

As the leader of this endeavor, our team embarked on an insightful journey, leveraging computer vision to meticulously analyze the behavior of customers upon entering a bustling superstore. Employing gender and age estimation techniques at the entrance, coupled with intricate footprint tracking throughout their visit, we empowered store owners to amplify sales and enhance overall user experiences.

2. Cutting-edge Chatbot for Proveo Health's Drug Database

Collaborating with Proveo Health, a distinguished US-based drug information company, our team engineered a sophisticated chatbot poised to revolutionize customer support. Focused on drug-related inquiries, users can seamlessly access dosage details, precautions, and side effects from their extensive database. This impactful solution harnessed Angular for intuitive front-end design and harnessed NLP and ML capabilities.

3. Deep Learning-Based Pipeline for Alzheimer's Disease Diagnosis

Developed a deep learning pipeline for the **medical** diagnosis and stratification of Alzheimer's Disease (AD) stages, contributing to advancements in healthcare. Utilized a shallow CNN architecture on 2D T1-weighted MRI brain images to improve medical imaging analysis. The project encompassed both global classification (normal vs. Mild Cognitive Impairment (MCI) vs. AD) and local classification, further stratifying MCI into Very Mild Dementia (VMD), Mild Dementia (MD), and Moderate Dementia (MoD).

4. SunAI - Empowering Solar Companies through IoT and ML

In a trailblazing venture, our team harnessed the potential of IoT and ML to introduce SunAI, a game-changing product tailored for the solar industry. Merging predictive analysis with optimized power scheduling, we facilitated solar panel production predictions and accurate electricity bill projections. Furthermore, we pioneered an end-to-end management system, equipping solar companies to seamlessly interact with both consumers and staff.

5. Amazon Product Ranking Revolution via Scraping and ML

Championing the realm of e-commerce, our project dynamically ranked diverse Amazon products through a fusion of scraping and machine learning prowess. Harnessing sentiment analysis via NLP techniques, we meticulously evaluated user reviews, while also discerning genuine feedback from deceptive ones. The end result was an ingenious web application, empowering users with invaluable insights.

6. IoT-Powered Pill Dispenser Revolutionizing Healthcare

Leading an innovation-driven team, we engineered a groundbreaking IoT-driven pill dispenser, redefining patient care. Through Bluetooth connectivity and global Internet access, caretakers gained unprecedented control, scheduling medications and remotely monitoring patient dose adherence. This pivotal solution not only enhances patient well-being but also streamlines caregiver responsibilities.

06/2016 – 07/2017 Riyadh, Saudi Arabia

SENIOR MACHINE LEARNING ENGINEER 4C GATE

1. Patients waiting time prediction algorithm

I developed an algorithm to forecast patients waiting time in different stages, like vital sign, doctor triage, pharmacies etc. I used different feature engineering and regression techniques to predict the waiting time in minutes.

2. MVP: Face Verification with OCR on Iqama/ID Cards for Insurance Companies

Developed a demo system for face verification based on ID cards using OpenCV. Employed advanced feature extraction techniques and classifiers to ensure accurate face matching. Additionally, designed an OCR algorithm to extract key details from Iqama cards, including the name, Iqama number, date of birth, and expiry date. The solution was aimed at preventing unauthorized use of insurance cards, enhancing security and compliance within the insurance industry.

06/2015 – 05/2016 Riyadh, Saudi Arabia

MACHINE LEARNING SPECIALIST EZAH COMPANY FOR TECHNOLOGIES

1. Developed an employee management system with facial recognition-based attendance system. I used an IP cam for taking images and then used machine learning techniques to classify the faces. I developed this system using C#.
2. Developed a vehicle management system based on number plate recognition. Time of entry and exit were recorded in our database. I used Canon IP cameras for taking video and then machine learning and image processing techniques to recognize numbers.
3. Lead a team for installation of canon IP cameras in Riyadh air base stadium.
4. Lead a team for installation and customer support of Zius keyless door security system. All doors were linked to SQL server.
5. Lead a team for troubleshooting of X-ray scanner in Continental hotel Riyadh.
6. Presentation of different security and surveillance technologies to clients.

04/2012 – 05/2015 Islamabad, Pakistan

AI ENGINEER EVEATI PVT. LTD.

1. Suspicious activities detection in CCTV using computer vision techniques

Algorithm was developed to detect suspicious activities from CCTV cameras footage. Models were trained to detect activities like running, climbing a wall, and crawling. Developed algorithm was handed over to an organization to integrate with their security and surveillance product.

2. Stock Prediction using newspaper

Aim of this project was to develop an algorithm that would help in predicting the future stock volumes (closing daily official volume for each trading day) of three leading companies (IBM, NETFLIX, IBM) by analyzing their previous stock data and newspaper headlines. Purpose was to foretell the stock market trends by analyzing the financial knowledge found online and then by using machine learning to interpret the text of news headlines.

3. Electricity Load Scheduling

This project encompasses the optimization of scheduling problems of home appliances. We know that any modern house is equipped with a vast array of electronic gadgets, but we also know that power supplier companies charge their consumers with an electricity tariff that fluctuates over 24 hours. We have proposed an algorithm that ensures that minimum cost is incurred by the consumers.

4. Radical Distribution Network Reconfiguration Using optimization techniques

Service restoration is an important aspect of power system design that caters to the restoration of power to unfaulty areas under blackout after an emergency condition. Once a fault takes place, the large number of customers in the blackout area relies upon the effectiveness of the load restoration mechanism. There are typically a large number of feeders with an even larger number of switches in a distribution system and it's not humanly possible to restore the service to an out-of-service area solely based on past experiences, that's why deep learning and heuristic optimization were deployed for the purpose.

EDUCATION AND TRAINING

02/2008 – 01/2012 Islamabad, Pakistan

BACHELOR IN ELECTRICAL ENGINEERING Comsats University, Islamabad

Islamabad, Pakistan

MASTER IN ARTIFICIAL INTELLIGENCE Air University

ONLINE COURSES

- Introduction to Deep Learning by National Research University Higher School of Economics
- Natural Language Processing by National Research University Higher School of Economics
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning by
- deeplearning.ai Neural Networks and Deep Learning by Deeplearning.ai
- Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization by Deeplearning.ai
- Structuring Machine Learning Projects by Deeplearning.ai
- Convolutional Neural Networks by Deeplearning.ai
- Intro to machine learning by Udacity
- Machine Learning for Time Series Data in Python by Datacamp
- Deep Learning in Python by Datacamp
- Introduction to SQL by Datacamp
- Joining Data in SQL by Datacamp

● **LANGUAGE SKILLS**

Mother tongue(s): **URDU**
Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C1	C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **DIGITAL SKILLS**

Deep learning | Machine learning | Computer vision | Natural Language Processing (NLP) | Predictive Analytics | Timeseries forecasting | Cloud computing | Project management | Linux | Keras | Tensorflow | Pandas | Matplotlib | Scikit-Learn | Git | Numpy | OpenCV | PyTorch | FLask | Django | Cloud platform: AWS, GCP etc. | Large Language Models | Amazon Web Services (Lambda, EC2, API Gateway, SageMaker, DynamoDB, RDS) | Nvidia Deepstream

● **PUBLICATIONS**

2024
[Enhancing Human Activity Recognition through Integrated Multimodal Analysis: A Focus on RGB Imaging, Skeletal Tracking, and Pose Estimation](#)

2017
[Action recognition by silhouette fusion](#)

Proceedings of the 6th International Conference on Software and Computer Applications (ACM).

2017
[A novel approach for feature extraction from RGB-D data](#)

Journal of Multidisciplinary Engineering Science Studies (JMESS) ISSN: 2458-925X Vol. 3 Issue 4

● **OTHER SKILLS**

Hardware

Arduino, Bluetooth, GSM, GPRS, RaspberryPi, PSoc, Infrared and ultrasonic sensors, Wifi Modules

Interests

Big Data, Databases, MLOps, Devops, Distributed computing